Ontology-Guided Knowledge Transfer and Class-Adaptive Post-Processing for Query-Based Sound Event Detection⋆

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ABSTRACT

Query-based sound event detection (SED) uses class queries to condition event recognition and temporal localization. Recent advanced frameworks, represented by the Detect Any Sound Model (DASM), achieve state-of-the-art performance on benchmarks like AudioSet by integrating class queries, cross-modal decoding, and frame-level mask prediction. However, in the practical audio-query setting, DASM faces two key challenges when detecting fine-grained and long-tailed sound events: first, query representations for rare subclasses suffer from data scarcity, background co-occurrence, and prototype noise; second, its reliance on a single, global median-filtering window ignores the substantial temporal scale heterogeneity across different event classes. To address these limitations, we propose an enhanced audio-query framework for target classes with training data. Specifically, Ontology-Guided Consistency Learning (OCL) regularizes refined queries via selected child-parent relations, complemented by Class-Wise Adaptive Weighting (CAW) that dynamically scales this regularization based on child-to-parent duration ratios. Furthermore, Class-Adaptive Median Filtering (CAMF) estimates class-specific smoothing windows from training durations using knee-based subset selection and a trimean estimator. Extensive experiments on AudioSet demonstrate that our method increases the overall Polyphonic Sound Detection Score (PSDS) from 49.73% to 51.26% and rare-class PSDS from 37.50% to 40.68%. These results confirm that our approach effectively strengthens long-tailed query representations while mitigating temporal scale mismatches in post-processing. Source code and audio demos are available at https://github.com/Hxx-source/ontology-guided-sed.

1. Introduction

Sound event detection (SED) recognizes acoustic event classes and estimates their onset and offset times, with system performance commonly summarized by standardized measures such as the Polyphonic Sound Detection Score [1–3]. It supports environmental perception, intelligent monitoring, multimedia retrieval, and context-aware audio understanding. The Detection and Classification of Acoustic Scenes and Events (DCASE) challenges have accelerated progress by providing shared datasets, evaluation protocols, and reproducible benchmarks for comparing SED systems [4, 5].

Conventional closed-set SED assumes a fixed label inventory shared by training and evaluation. Çakır et al. introduced a convolutional recurrent neural network (CRNN) that combines local spectro-temporal feature extraction with recurrent sequence modeling [2]. Nam et al. proposed frequency-dynamic convolution to adapt acoustic pattern modeling across frequency bands [6]. Kong et al. developed PANNs to transfer large-scale AudioSet pretraining to downstream audio tasks [7]. Transformer-based models then strengthened long-range modeling: Gong et al. proposed AST and its self-supervised extension SSAST [8, 9], while Chen et al. introduced HTS-AT with a hierarchical token-semantic architecture [10]. These methods substantially improve SED representations, but their class-specific output layers remain tied to the label space defined during training.

Open-vocabulary and query-based approaches instead specify the target through text or acoustic prompts. Elizalde et al. introduced CLAP to learn a shared audio–language embedding space [11], and Wu et al. developed LAION-CLAP using large-scale audio-text pretraining [12]. For target sound detection, Yang et al. proposed RaDur, which uses reference audio and duration-robust modeling to identify a specified sound [13]. Hai et al. presented FlexSED, a text-query-driven framework for localizing queried sound events [14]. These studies show that query conditioning can decouple target specification from a fixed classification head and support a shared detector across different targets.

Among these query-conditioned frameworks, the Detect Any Sound Model (DASM) [15] achieves state-of-the-art performance on benchmarks such as AudioSet by combining class queries, cross-modal decoding, and temporal context modeling. However, when applied to fine-grained and long-tailed sound events, DASM suffers from two major practical limitations: rare-subclass queries are highly susceptible to data scarcity, background co-occurrence, and prototype noise, while its reliance on a uniform global median-filtering window ignores the substantial temporal scale differences across event classes.

To address these limitations, we focus on the audio-query setting where class queries are built directly from training audio clips, and propose an enhanced framework

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that improves both query representations and temporal post-processing without altering the core backbone architecture. Specifically, we leverage ontology-derived hierarchical knowledge to regularize data-scarce subclass queries against parent classes, preventing representation degradation caused by noise and scarcity. In addition, we replace the global post-processing strategy with an automated class-adaptive filtering mechanism that tailors smoothing windows to individual event duration distributions. The main contributions of this work are summarized as follows:

- We propose a reliable child-parent relation selection strategy, which jointly considers the duration ratio, parent-class support, and cosine similarity between audio queries to identify ontology relations suitable for knowledge transfer.
- We introduce Ontology-Guided Consistency Learning (OCL) complemented by Class-Wise Adaptive Weighting (CAW), which stabilizes rare-subclass queries by transferring parent-level knowledge weighted by relative duration support.
- We propose Class-Adaptive Median Filtering (CAMF), an automatic post-processing strategy that estimates class-specific smoothing windows via knee-based subset selection and trimean estimation, resolving temporal scale mismatches across heterogeneous sound events.
- Extensive experiments on AudioSet validate the effectiveness of our framework, achieving consistent improvements in overall PSDS (from 49.73% to 51.26%) and rare-class PSDS (from 37.50% to 40.68%).

2. Related Work

2.1. Query-based Sound Event Detection

Contrastive audio–language pretraining provides shared representations for audio and text. CLAP [11] and LAION-CLAP [12] learn such representations at scale, while AudioCLIP and Pengi explore related audio–language formulations [16, 17]. Audio-caption resources and audio-language models broaden the available supervision and task formulations [18–21]. For SED, DASM formulates detection as query-conditioned frame-level retrieval using text or audio queries and a dual-stream decoder [15]; FlexSED also studies text-query-driven detection [14]. Language-conditioned audio analysis further includes sound separation and general audio understanding [22–24]. These studies demonstrate the utility of query-based modeling. The present work focuses specifically on improving DASM's audio-query representations for a predefined target set rather than evaluating unseen-class generalization.

Query-based audio analysis also relies on strong acoustic encoders. PANNs demonstrated the value of large-scale convolutional pretraining for downstream recognition [7], while frequency-adaptive convolutions improve spectro-temporal pattern modeling [6]. Transformer-based encoders subsequently improved long-range modeling: AST introduced a fully attention-based architecture [8], SSAST extended it with self-supervised pretraining [9], PaSST used patchout for efficiency and regularization [25], and HTS-AT introduced hierarchical token-semantic modeling [10]. Audio-MAE, BEATs, ATST, and CAV-MAE provide additional self-supervised or audiovisual representations [26–29]. Attention-based architectures have also been developed for joint sound event localization and detection [30]. These advances strengthen the features used by query-based detectors, but do not directly address ontology-linked query regularization or class-dependent post-processing.

2.2. Ontology and Hierarchical Learning

AudioSet organizes hundreds of sound classes in a hierarchical ontology and provides a natural basis for large-scale audio recognition [31]. Sun and Ghaffarzadegan proposed an ontology-aware framework for audio event classification that models label dependencies through a feed-forward ontology layer and a graph convolutional network [32]. More recent hierarchical label-propagation approaches identify parent–child inconsistencies and propagate labels upward through the ontology [33]. These studies use ontology structure for relationship modeling and label correction, whereas the present work constrains audio-conditioned query representations without changing the evaluated label set.

2.3. Post-Processing and Temporal Modeling

Temporal modeling and post-processing are also central to SED performance. Conventional CRNN-based systems and DCASE Task 4 systems formulate SED as frame-level multilabel prediction and apply thresholding and temporal filtering to obtain event boundaries [2, 5]. Post-processing can substantially affect event-level detection and system comparison [34]. PSDS therefore evaluates systems over operating points using intersection-based event matching [1]. Existing methods often use a global or empirically selected window, while adaptive feature smoothing operates inside a model rather than on output scores [35]. With hundreds of target classes and widely varying durations, a fixed median-filtering window cannot adequately serve both short impulsive and long sustained events.

3. Revisit the DASM Framework

As shown in the lower part of Fig. 1, DASM is a query-based SED framework, and the blue components indicate the original baseline forward path. Given an input audio clip, an HTS-AT backbone and a convolutional encoding branch first extract encoded audio features. In parallel, an audio query is mapped to a query embedding that specifies the target sound class. The query embedding and encoded audio features are then passed to the dual-stream DASM decoder. Its cross-modal event decoder models correspondence between query and recording features and produces audio-conditioned refined query vectors, while the context network models temporal dependencies. The outputs are fused to generate

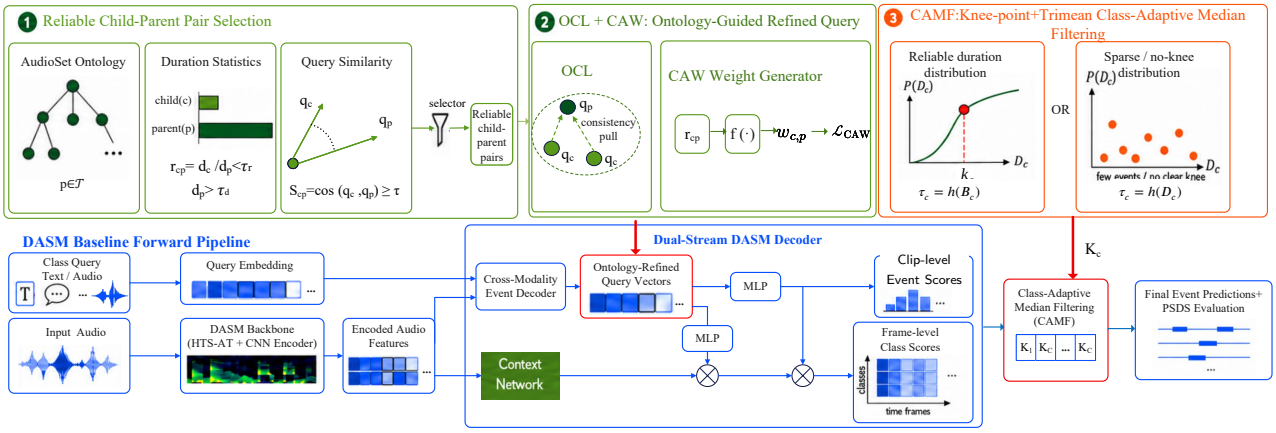


Figure 1: Framework of the proposed method. The lower pathway shows the DASM backbone, query decoder, and frame-score computation. The upper modules show relation selection, OCL+CAW regularization of projected refined queries, and CAMF duration estimation. OCL+CAW acts during training; CAMF smooths frame scores at inference.

clip-level predictions and frame-level class scores, which are converted into event detections by post-processing.

As shown in [15], DASM has achieved the state-of-the-art performance in query-based SED. In this work, we adopt its audio-query setting as our primary baseline and conduct in-depth research to further advance its capabilities. Specifically, we extend DASM from three aspects: introducing ontology-guided knowledge transfer for rare-class queries, designing dynamic adaptive weighting for hierarchical regularization, and developing class-adaptive post-processing for temporal scale alignment.

4. Proposed Method

The proposed method retains the audio-query DASM forward architecture and enhances it at two key stages: query representation learning during training and class-adaptive post-processing during inference. First, reliable child-parent relations are selected from the sound event ontology. A consistency constraint with adaptive weighting is then applied to the refined query vectors produced by the cross-modal event decoder, enabling data-scarce child classes to leverage representations from better-supported parent classes. Finally, class-specific median-filtering windows are estimated from training-duration statistics to optimize post-processing for heterogeneous event scales. Details are presented as follows.

4.1. Ontology-Guided Consistency Learning

The Ontology-Guided Consistency Learning (OCL) is designed to exploit reliable parent-class information from the sound ontology to improve query representation learning for data-scarce child classes. Rather than modifying the DASM backbone or changing the class labels, the consistency constraint is imposed on the refined query vectors output by the cross-modal event decoder. These representations have already incorporated the context of the input audio and therefore reflect the model's discriminative interpretation of the target class more directly than the original static query vectors. As shown in the upper part of Fig. 1, a lightweight projector maps the refined child and parent query vectors into a shared semantic space, where their cosine similarity is measured. For selected reliable child-parent relations, a margin-based consistency loss encourages the child representation to approach the semantic neighborhood of its parent. No additional penalty is imposed once the cosine similarity exceeds the prescribed margin. This relation-level objective differs from teacher–student prediction consistency [36]: it constrains ontology-linked query representations rather than predictions for perturbed inputs. OCL therefore complements, rather than replaces, the original DASM supervision.

4.1.1. Ontology-Based Candidate Selection

To facilitate effective knowledge transfer, we define a selection strategy that identifies robust child-parent relationships within the AudioSet hierarchy, ensuring that parent classes serve as stable semantic anchors for data-scarce child classes. As shown in Fig.1 (1), let $\mathcal{T}$ be the set of all direct parent-child pairs in the ontology. We filter $\mathcal{T}$ to form a reliable subset $\mathcal{P} \subseteq \mathcal{T}$ by considering temporal distribution statistics and semantic alignment without recourse to validation data.

For each pair $(c, p) \in \mathcal{T}$, we denote d_c and d_p as the cumulative training durations of the child and parent classes, respectively. We define the duration ratio $r_{c,p} = d_c/d_p$ and the cosine similarity between pretrained audio query embeddings as $s_{c,p} = \cos(q_c, q_p)$. A relation is included in $\mathcal{P}$ if it satisfies:

1) **Sufficient Parent Support:** $d_p > \tau_d$, with $\tau_d = 3$ hours. This practical duration floor favors parents with broader acoustic support over parents that are themselves data-scarce.
2) **Prototype Alignment:** $s_{c,p} \in [\tau_{\min}, \tau_{\max}] = [0.85, 0.98]$. The lower bound retains acoustically aligned prototypes; the upper bound excludes nearly identical prototypes for which further alignment may be redundant.
3) **Target Scarcity:** $r_{c,p} < \tau_r$, with the fixed threshold $\tau_r = 0.2$. This requires more than five times as much

parent duration as child duration and targets a substantial support imbalance.

The three criteria address complementary properties: absolute parent support, relative child scarcity, and acoustic agreement. Their intersection defines candidate relations but does not guarantee semantic equivalence or eliminate negative transfer. The thresholded candidates are ranked by duration ratio, parent support, and similarity reliability; the top 60 relations, covering 59 child classes, are retained. This budget provides coverage for the selected low-resource children while preventing the auxiliary objective from accumulating many lower-ranked, potentially noisy relations. The thresholds and relation budget are fixed design choices rather than values established as optimal by a sensitivity study.

4.1.2. Consistency Loss

Given the reliable child-parent relation set $\mathcal{P}$ obtained by the ontology-based candidate selection strategy, we impose consistency regularization only on these filtered relations to facilitate semantic knowledge transfer from data-rich parent classes to long-tailed child classes.

As illustrated in Fig.1 (2), for each selected pair $(c, p) \in \mathcal{P}$, let h_c and h_p denote the refined query vectors generated by the DASM cross-modal event decoder. A learnable projector $g(\cdot)$ is employed to map them into a shared semantic space:

$$z_c = \text{normalize}\left(g\left(h_c\right)\right), \quad z_p = \text{normalize}\left(g\left(h_p\right)\right) \tag{1}$$

To encourage semantic alignment between child and parent classes, we define the ontology consistency loss ($\mathcal{L}_{\text{OCL}}$) as:

$$\mathcal{L}_{\text{OCL}} = \frac{1}{|\mathcal{P}|} \sum_{(c,p)\in\mathcal{P}} \max\left(0, m - \cos\left(z_c, z_p\right)\right) \tag{2}$$

where $\mathcal{P}$ denotes the selected child–parent pairs, $\cos(\cdot,\cdot) \in [-1, 1]$, and $m = 0.7$ is the similarity margin. L2 normalization makes the loss depend on vector direction rather than magnitude. The moderate margin encourages meaningful agreement in the projected space without forcing child and parent representations to become nearly identical. Below the margin, the hinge penalty decreases linearly with similarity; at or above 0.7, it is zero. Thus, the loss encourages sufficient agreement while preserving child-specific variation. The prototype-selection interval $[0.85, 0.98]$ and the projected-query margin m act on different representations and serve different purposes.

During optimization, the refined parent query is detached in the consistency branch, serving as a stable teacher anchor. Consequently, gradients from $\mathcal{L}_{\text{OCL}}$ primarily update the child representations, while parent-related parameters continue to be optimized through the original DASM supervision. This design allows rare child classes to benefit from robust semantic guidance while avoiding over-smoothing and preserving fine-grained discriminability.

The final training objective combines the original DASM loss with the proposed ontology consistency regularization:

$$\mathcal{L} = \mathcal{L}_{\text{DASM}} + \lambda\mathcal{L}_{\text{OCL}} \tag{3}$$

where λ controls the contribution of the ontology consistency loss.

4.2. Class-Wise Adaptive Weighting

While the basic $\mathcal{L}_{\text{OCL}}$ treats all relations in $\mathcal{P}$ equally, child classes differ significantly in their dependence on parent knowledge. Child classes with extremely limited training data are more susceptible to sampling bias and require stronger guidance, whereas classes with sufficient training support risk being over-smoothed if the consistency constraint is too rigid. To address this, we introduce Class-Wise Adaptive Weighting (CAW), which assigns a dynamic importance weight to each child-parent relation based on the previously defined ratio $r_{c,p}$.

We define the relation weight $w_{c,p}$ as a non-increasing function of $r_{c,p}$:

$$w_{c,p} = \text{clip}\left(\frac{\log\left(1/r_{c,p}\right)}{\log(1000)}, 0, 1\right) \tag{4}$$

The logarithm compresses support differences across orders of magnitude. The denominator $\log(1000)$ defines a reference imbalance of 1000:1: ratios 10^{-3}, 10^{-2}, and 10^{-1} receive weights 1, $2/3$, and $1/3$. Clipping caps the weight at 1 for $r_{c,p} \le 10^{-3}$. Under the selection condition $0 < r_{c,p} < 0.2$, admissible weights lie in approximately $(0.233, 1]$; thus, selected relations retain positive weights rather than decaying to zero.

Integrating this weighting scheme into our consistency regularization, the weighted ontology consistency loss is defined as:

$$\mathcal{L}_{\text{CAW}} = \frac{1}{|\mathcal{P}|} \sum_{(c,p)\in\mathcal{P}} w_{c,p} \cdot \max\left(0, m - \cos\left(z_c, z_p\right)\right) \tag{5}$$

With CAW enabled, $\mathcal{L}_{\text{CAW}}$ replaces $\mathcal{L}_{\text{OCL}}$ in the final objective. The weights are computed from training durations and remain fixed during optimization. Because the denominator is $|\mathcal{P}|$ rather than $\sum w_{c,p}$, CAW changes both relative pair contributions and the aggregate auxiliary-loss scale. We set $\lambda = 0.1$ to keep ontology consistency as an auxiliary regularizer rather than allowing it to dominate the original DASM supervision. This coefficient controls the loss scale but does not imply a ten-to-one ratio between supervised and auxiliary gradients.

4.3. Class-Adaptive Median Filtering

Post-processing frame-level scores into event boundaries typically involves applying a fixed median-filtering window to all classes. However, AudioSet-Strong encompasses 407 classes with heterogeneous temporal structures, ranging from transient impulsive events to sustained sounds. A rigid window inevitably suppresses responses for short events or leads to fragmented predictions for sustained ones. To address this, we propose Class-Adaptive Median Filtering (CAMF), which derives an independent, data-driven window length K_c for each class based on its training-set event-duration distribution.

For each class c, we collect all training-event durations and sort them in ascending order as $D_c = \{d_{(1)}, \ldots, d_{(N)}\}$. Knee-point detection identifies a change in curve shape [37]; here, a knee at index k_c partitions the duration sequence. When the knee passes the reliability check, we estimate the class-specific temporal scale τ_c from the subset B_c defined below; otherwise, we use the full set D_c:

$$\tau_c = \begin{cases} h(B_c), & \text{if knee point is reliable} \\ h(D_c), & \text{otherwise} \end{cases} \tag{6}$$

where the dominant subset B_c and the relative knee position $\rho_c = k_c/N$ are defined as:

$$B_c = \begin{cases} \{d_{(k_c+1)}, \ldots, d_{(N)}\}, & \rho_c < 0.5 \\ \{d_{(1)}, \ldots, d_{(k_c)}\}, & \rho_c \geq 0.5 \end{cases} \tag{7}$$

The threshold 0.5 selects the larger side of the knee partition, avoiding a temporal estimate based on a small tail of the duration distribution. If no reliable knee is found, using the full set D_c prevents an unstable split from determining the window. The trimean fallback $h(\cdot)$ is defined as:

$$h(D_c) = \frac{Q_1(D_c) + 2Q_2(D_c) + Q_3(D_c)}{4} \tag{8}$$

where Q_1, Q_2, Q_3 denote the first, second (median), and third quartiles of D_c, respectively. This estimator provides a robust central tendency that is more resilient to outliers than the mean while incorporating more distributional information than the median alone.

Finally, we derive the class-specific window size K_c by applying a scaling factor $\gamma = 1/3$, clipping the result, and rounding to the nearest odd integer:

$$K_c = \text{odd}\left(\text{clip}\left(\gamma\tau_c, K_{\min}, K_{\max}\right)\right) \tag{9}$$

with $K_{\min} = 3$ and $K_{\max} = 101$ output frames. Durations are converted to output-frame units before scaling. The factor $\gamma = 1/3$ keeps the unclipped window shorter than the representative duration; the lower bound excludes identity filtering, the upper bound limits excessive smoothing, and odd sizes provide a unique central sample. At inference, each class c uses its corresponding window K_c to smooth its frame-level score y_c:

$$\hat{y}_c = \text{MedFilt}(y_c, K_c) \tag{10}$$

Fig.2 shows the CAMF window-size distribution across the 407 AudioSet-Strong classes. The fixed 16-frame window follows the baseline DASM post-processing configuration and therefore serves as the reference rather than a value retuned on the validation set. The class-specific windows span a broad range from 3 to 101 frames, with a mean of 36.5 frames. Most short-duration classes receive small windows, whereas many sustained classes receive larger windows, some reaching the maximum of 101 frames. This distribution confirms both the pronounced variation in event duration across classes and the limitation of a global median-filtering window. CAMF uses only training-set duration statistics, does not access validation- or test-set information, and introduces no additional model parameters. It can therefore be applied as a lightweight post-processing module to the frame-level outputs.

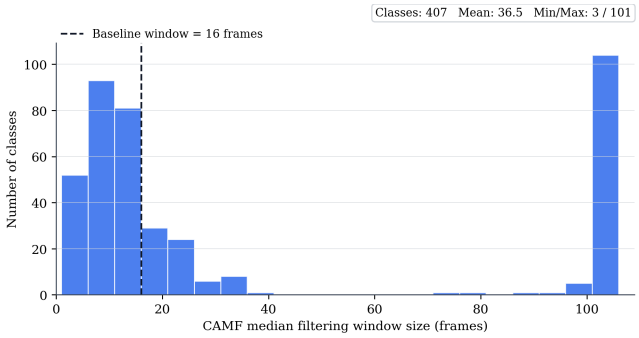


Figure 2: CAMF window-size distribution for the 407 target classes on AudioSet-Strong. The dashed line denotes the fixed 16-frame baseline window.

5. Experimental Setup

5.1. Dataset

Experiments use AudioSet-Strong, the temporally annotated portion of AudioSet [31, 38]. Following DASM [15], the detector uses a predefined set of 407 target classes from the AudioSet ontology. Strongly labeled training events supervise the detector and provide the class counts, cumulative durations, child-to-parent duration ratios, and event-duration samples used by the proposed components.

Overall and class-wise metrics are reported on the AudioSet-Strong validation set. Because not all 407 classes contain valid annotations for class-wise evaluation in the current validation split, 381 classes participate in the class-level analysis. To investigate long-tailed performance, we use the training event counts together with the existing rare-class partition to designate 81 low-resource classes as rare and the remaining classes as common. Overall metrics are computed over all valid classes, whereas rare- and common-class metrics are calculated over their corresponding subsets. This partition makes it possible to determine whether ontology-guided transfer genuinely improves data-scarce classes rather than merely increasing an aggregate score dominated by common classes.

5.2. Model Configuration

The baseline is the HTS-AT [10] audio-query variant of DASM. The audio encoder uses an HTS-AT backbone together with a convolutional branch to extract local time-frequency features. Class queries are represented by audio-query vectors whose dimensionality is aligned with the hidden space of the DASM decoder. The original dual-stream DASM architecture is retained: the cross-modal event decoder and context network produce refined query vectors, clip-level predictions, and frame-level class scores. The supervised DASM objective combines a strongly labeled SED branch

Table 1
Overall PSDS results on AudioSet-Strong (%).

ID	Method	PSDS	Common	Rare
A0	Baseline (DASM)	49.73	53.21	37.50
A1	+OCL	50.01	53.26	38.49
A2	+OCL+CAW	50.22	53.33	39.24
A3	+OCL+CAW+CAMF	**51.26**	**54.27**	**40.68**

Table 2
Comparison with other sound event detection methods on AudioSet-Strong (%).

Method	PSDS	Common	Rare
HTS-SED [15]	44.0	47.3	31.8
LAION-CLAP [12]	38.1	41.4	25.7
MGA-CLAP [41]	40.9	44.2	28.8
DASM [15]	49.7	53.2	37.5
Ours	51.3	54.3	40.7

loss for frame-level localization with an audio-tagging branch loss that constrains clip-level class presence.

Performance is evaluated using the Polyphonic Sound Detection Score (PSDS), which summarizes detection performance across operating points using intersection-based event matching [1, 39]. All reported PSDS values are multiplied by 100; differences are stated in points on this scale rather than relative percentages. Post-processing is part of the evaluation configuration because temporal smoothing changes the intervals supplied to the metric [34]. ROC and precision–recall analyses characterize different aspects of imbalanced binary classification [40]; PSDS remains the primary event-detection measure.

6. Results and Discussion

6.1. Overall Results on AudioSet-Strong

Table 1 summarizes the ablation study results. The A0 baseline (DASM) yields an overall PSDS of 49.73%, yet the significant performance gap between common (53.21%) and rare classes (37.50%) underscores the persistent challenge of long-tailed distribution. Integrating OCL (A1) improves the rare-class PSDS to 38.49%, demonstrating that hierarchical consistency provides effective semantic guidance for data-scarce child classes. Further applying CAW (A2) boosts this to 39.24% by dynamically prioritizing consistency constraints for the most data-deprived classes, thereby maximizing transfer efficiency while preventing over-regularization on well-supported categories.

The final configuration, A3, incorporates CAMF and achieves a superior overall PSDS of 51.26%. While OCL and CAW specifically alleviate representation-level bottlenecks for rare classes, CAMF addresses temporal localization errors by customizing inference-time windows to match individual class-specific event dynamics. Consequently, CAMF yields performance gains across both common and rare subsets. Together, these results validate that our approach successfully mitigates both the semantic representation instability of long-tailed classes and the temporal mismatch inherent in fixed-window post-processing.

6.2. Comparison with Other SED Methods

Table 2 compares the complete method with HTS-SED [15], LAION-CLAP [12], MGA-CLAP [41], and DASM [15] on AS-full. By retaining the DASM backbone and decoder while improving query regularization and temporal smoothing, our method achieves the highest scores in all three categories: 51.3% overall, 54.3% on common classes, and 40.7% on rare classes. Compared with DASM, these results correspond to gains of 1.6, 1.1, and 3.2 points, respectively. The larger improvement on rare classes highlights the effectiveness of ontology-guided knowledge transfer and class-adaptive post-processing for data-scarce sound events.

6.3. Query Discriminability

Figure 3 compares the target-versus-rest discriminability of refined queries for six representative classes. ROC curves summarize the trade-off between true-positive and false-positive rates as the decision threshold changes. If target and non-target clips are well separated, high sensitivity is maintained with few false alarms, moving the curve toward the upper-left corner; greater overlap moves it toward the diagonal. Thus, higher AUROC indicates stronger target-versus-rest discriminability [42]. Gray, blue, and red denote the baseline, OCL, and OCL+CAW, respectively.

Both OCL and OCL+CAW improve AUROC over the baseline in all six classes. In particular, OCL+CAW yields gains of 9.70 points for *Sniff* and 8.74 points for *Gears*. These consistent improvements show that ontology-guided regularization produces more discriminative refined queries and strengthens target-related information for downstream detection.

6.4. Rare-Class Analysis

To identify the sources of the improvement, Table 3 lists the 10 rare child classes with the largest PSDS gains. These classes exhibit pronounced long-tailed characteristics: the child-class training durations are short, whereas their parent classes have substantially greater support. For example, *Squish* has only 0.027 hours of training data, while its parent class *Liquid* has 5.50 hours, yielding a child-to-parent duration ratio of 0.0050. In the final system, the PSDS of *Squish* increases from 15.62% to 30.84%, a gain of 15.22 percentage points. Similarly, the parent class *Wild animals* provides 10.27 hours of training data for *Snake*; *Smash, crash*, *Whack, thwack*, *Slap, smack*, and *Flap* all benefit from the better-supported parent class *Generic impact sounds*. These cases show that a parent class can effectively regularize a child query when the child is extremely data-scarce and the parent has strong statistical support.

A second pattern in Table 3 is that high-gain child-parent pairs generally exhibit strong but non-saturated audio-query similarity. The similarities of *Squish–Liquid*, *Snake–Wild animals*, and *Goat–Livestock* are around 0.90 or higher,

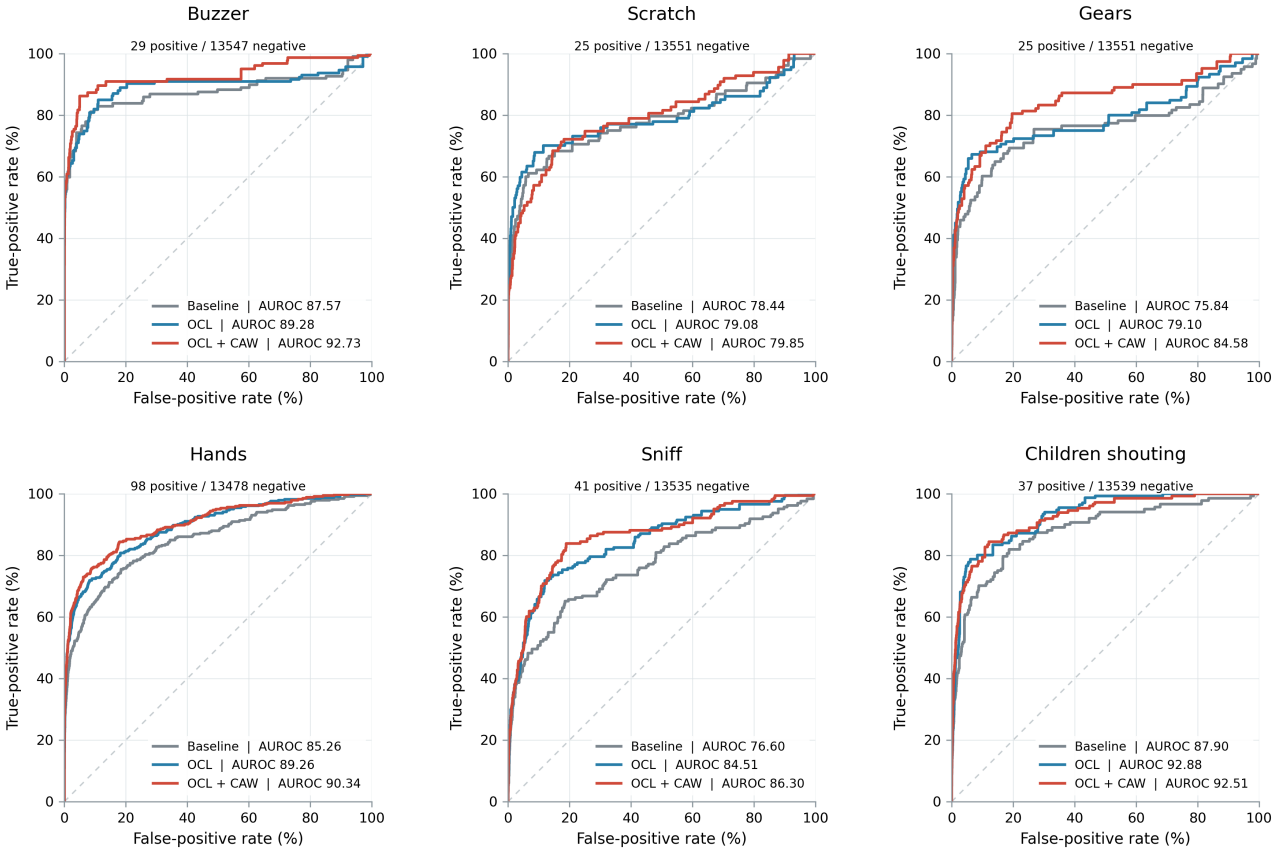


Figure 3: ROC curves comparing refined-query discriminability for six selected classes. Legends report AUROC (%) for the baseline, OCL, and OCL+CAW.

Table 3
Top 10 rare subclasses with the largest PSDS improvements.

Rank	Child	Parent	d_c (h)	d_p (h)	$r_{c,p}$	$s_{c,p}$	DASM	Ours	Δ
1	Squish	Liquid	0.027	5.50	0.0050	0.909	15.62	**30.84**	+15.22
2	Snake	Wild animals	0.071	10.27	0.0069	0.939	39.79	**50.27**	+10.48
3	Smash, crash	Generic impact sounds	0.011	11.01	0.0010	0.860	61.16	**69.53**	+8.37
4	Goat	Livestock, farm animals, working animals	0.097	3.32	0.0292	0.928	44.95	**52.97**	+8.02
5	Whack, thwack	Generic impact sounds	0.042	11.01	0.0038	0.865	22.04	**29.09**	+7.05
6	Buzzer	Alarm	0.072	5.81	0.0123	0.943	49.15	**56.08**	+6.93
7	Wheeze	Breathing	0.018	4.69	0.0037	0.949	31.67	**36.51**	+4.84
8	Snort	Breathing	0.031	4.69	0.0067	0.912	37.65	**42.30**	+4.65
9	Slap, smack	Generic impact sounds	0.054	11.01	0.0049	0.878	26.37	**30.86**	+4.49
10	Flap	Generic impact sounds	0.019	11.01	0.0017	0.875	10.16	**13.72**	+3.56

indicating substantial correspondence in the semantic and acoustic embedding spaces. Although pairs such as *Smash, crash–Generic impact sounds* and *Whack, thwack–Generic impact sounds* have slightly lower similarities, the much greater training coverage of the parent still provides a valuable transfer signal. This observation agrees with the candidate selection strategy: a reliable parent must not only be linked to the child in the ontology, but must also satisfy data-support and query-similarity criteria.

Figure 4(a) provides a complementary view over the full rare-class distribution. Rare classes are ordered on the horizontal axis by their baseline PSDS from low to high, so the left side corresponds to classes that are difficult for the original DASM and the right side to classes with stronger baseline performance. Gray points and the gray smoothed curve denote the baseline distribution, whereas blue, orange, and red points denote OCL, OCL+CAW, and OCL+CAW+CAMF, respectively. Relative to the baseline, red points lie above the gray trend more frequently in the

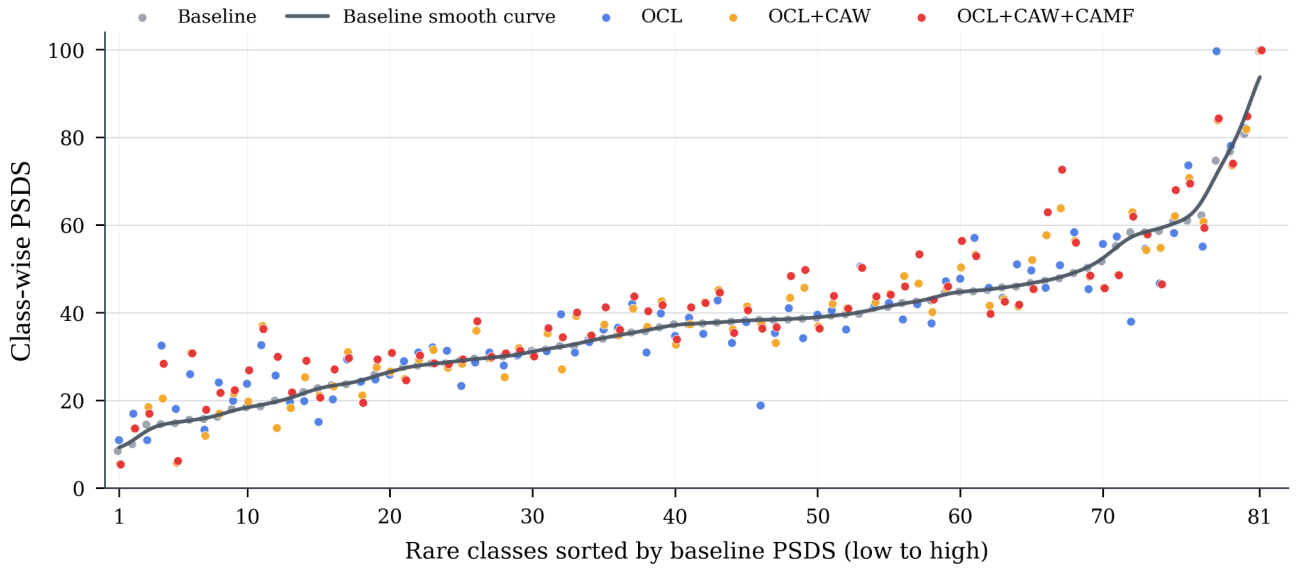


Figure 4(a): Class-wise PSDS for the 81 rare validation classes, ordered by baseline PSDS. Gray points denote the baseline; blue, orange, and red denote OCL, OCL+CAW, and OCL+CAW+CAMF. The dark curve is a smoothed baseline guide; improvements should be assessed against the paired baseline points.

low- and middle-score regions. The final method therefore improves class-wise PSDS across a broad set of rare classes rather than only a few isolated cases.

Improvements are less consistent in the high-baseline region, which is expected because these classes are already easier to detect and offer less headroom, while changes in the post-processing window can introduce trade-offs. Taken together, Table 3 and Fig. 4(a) support the central hypothesis: when child-class data are scarce, parent-class data are abundant, and their semantic and acoustic relationship is reliable, ontology-guided consistency learning provides an effective semantic anchor. Relation-specific weighting prioritizes child-parent pairs with greater transfer value, while class-adaptive post-processing contributes complementary gains through temporal modeling.

6.5. Common-Class Analysis

Although the proposed method primarily targets long-tailed rare classes, Table 4 shows that several common child classes also improve markedly. Using the same child-parent analysis as in Table 3, we exclude the rare classes and select the five common child classes with the largest PSDS gains. *Environmental noise* exhibits the largest gain, increasing from 27.96% to 41.47%, or 13.51 percentage points. Although this class is not assigned to the rare subset, it has only 0.152 hours of training data, whereas its parent class *Background noise* has 47.42 hours. The resulting duration ratio is only 0.0032, and the audio-query similarity is 0.941. Thus, a class categorized as common can still be severely underrepresented relative to its parent and can benefit from parent-guided consistency and relation-specific weighting.

Similarly, *Rodents, rats, mice* has 0.659 hours of training data, compared with 10.27 hours for its parent class *Wild animals*; its PSDS increases from 24.12% to 34.31%. *Synthetic singing–Singing* and *Camera–Mechanisms* exhibit the same pattern: the parent classes have greater training support and the child-parent query similarities remain high, enabling the parent queries to provide useful representation anchors. *Motorcycle* already has strong baseline performance, yet still improves from 89.17% to 94.51%, indicating that parent-class constraints and class-adaptive post-processing can further stabilize scores even for an easily detectable class.

These results demonstrate that the benefit of the proposed method is not determined solely by a binary partition into rare and common classes. It depends more directly on the support imbalance within a specific child-parent relation, query similarity, and event temporal structure. In other words, a common class may still be data-scarce relative to its parent. When the parent has substantially greater training support and the pair is reliably related in acoustic-semantic space, OCL and CAW can still provide effective transfer. Because CAMF adjusts post-processing according to the event-duration distribution of each class, it can also improve frame-level smoothing and event-boundary stability for common classes. Table 4 therefore confirms that the proposed method provides additional gains beyond the rare subset.

6.6. Effect of CAMF on Class Distributions

Figure 4(b) compares fixed-window post-processing with CAMF over all 381 validation classes and shows that CAMF affects a broad range of classes rather than only a small number of cases. Of the 381 classes, 239 improve, 133 decrease, and the remaining 9 are unchanged. The mean PSDS gain is 0.82 percentage points and the median gain is 0.27 percentage points, indicating an overall positive effect from class-adaptive post-processing.

Table 4
Top 5 common subclasses with the largest PSDS improvements.

Rank	Child	Parent	d_c (h)	d_p (h)	$r_{c,p}$	$s_{c,p}$	DASM	Ours	Δ
1	Environmental noise	Background noise	0.152	47.42	0.0032	0.941	27.96	**41.47**	+13.51
2	Rodents, rats, mice	Wild animals	0.659	10.27	0.0642	0.920	24.12	**34.31**	+10.19
3	Synthetic singing	Singing	0.475	12.97	0.0366	0.926	61.27	**68.95**	+7.68
4	Camera	Mechanisms	0.188	66.34	0.0028	0.879	30.12	**37.12**	+7.00
5	Motorcycle	Motor vehicle (road)	1.385	14.43	0.0960	0.940	89.17	**94.51**	+5.34

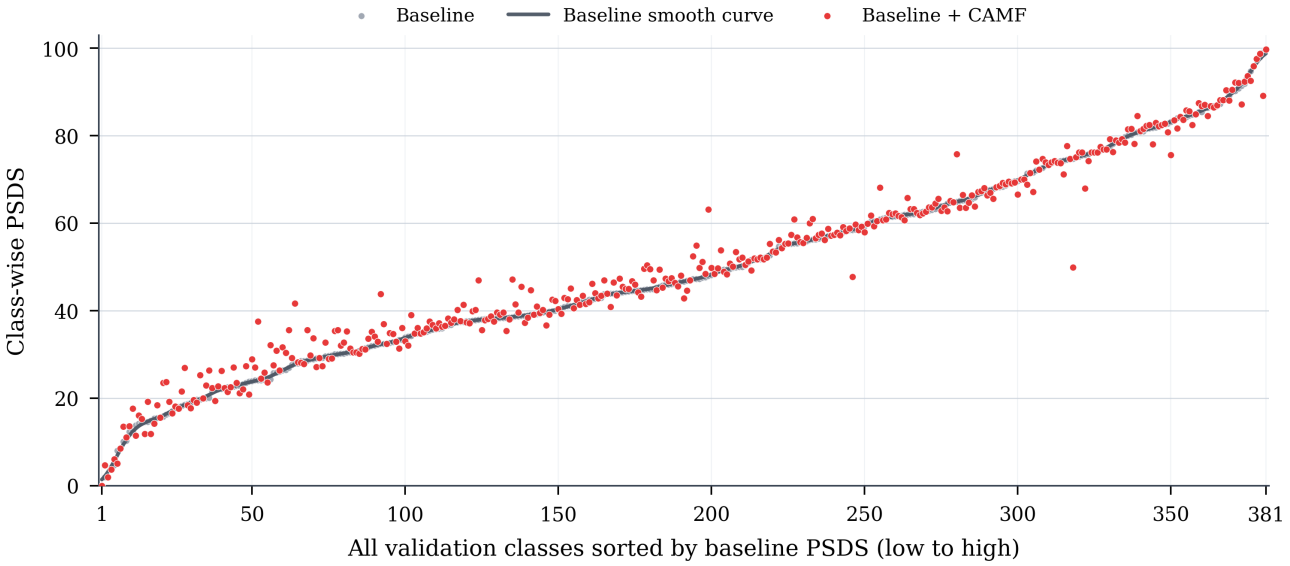


Figure 4(b): Class-wise PSDS for fixed baseline post-processing versus baseline+CAMF over all 381 valid validation classes. Classes are ordered by baseline PSDS and retain the same order after CAMF. Gray and red points denote the fixed-window baseline and CAMF; the dark curve is a smoothed baseline guide.

The gains are concentrated primarily in the low- and middle-baseline regions, where the fixed reference window is often poorly matched to class-specific temporal structure. An excessively large window can suppress transient responses from short events, whereas an overly small window can fragment sustained events. CAMF independently estimates a filtering window for each class from its training-set duration statistics and therefore better accommodates heterogeneous temporal patterns. This also explains why CAMF benefits both rare and common classes: it optimizes class-specific temporal post-processing rather than relying solely on class frequency.

CAMF does not improve every class monotonically. Performance can decrease for classes with sparse duration distributions, substantial annotation noise, or complex and variable event boundaries, for which duration statistics alone may not provide a sufficiently accurate window estimate. CAMF should therefore be understood as an improvement to the overall strategy for class-wise temporal smoothing rather than as a replacement for every class's optimal threshold or post-processing rule. Nevertheless, the results support the hypothesis that the evaluated classes exhibit heterogeneous duration structures and that class-adaptive median filtering can be more appropriate than a single fixed window. It improves overall PSDS without adding learnable parameters or tuning on the validation set.

7. Conclusion

This paper presented ontology-guided query regularization and class-adaptive post-processing for the audio-query variant of DASM. OCL constrains projected refined queries for selected child–parent relations, CAW weights each relation by its child-to-parent training-duration ratio, and CAMF derives class-specific smoothing windows from training-event durations. On the evaluated AudioSet-Strong split, the complete system improves overall PSDS from 49.73 to 51.26, rare-class PSDS from 37.50 to 40.68, and common-class PSDS from 53.21 to 54.27. The linear-probe results indicate more accessible target-presence information for the six illustrated classes, while the class-wise analyses show that gains vary across classes. These findings support the combined method within the evaluated predefined target set; they do not establish unseen-class generalization or transfer to other detector architectures.

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